

Major Project Synopsis

An AI-Based System for Cyberbullying Detection

A Comparative Study of Machine Learning and Deep Learning Approaches

Submitted by:

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1. Introduction

Social media has become an inseparable part of our daily lives. It connects people instantly across the globe, but unfortunately, it also provides space for negative behaviors like **cyberbullying**. Online harassment, abusive comments, and threatening messages have become a serious problem, often affecting the mental health and confidence of individuals.

Our project focuses on creating a **Cyberbullying Detection System** that can automatically recognize such harmful content. By using techniques from **Machine Learning (ML)**, **Deep Learning (DL)**, and **Natural Language Processing (NLP)**, we aim to detect abusive text in real time.

To make this solution practical and easy to use, we plan to build it as a **browser extension**. This way, users can get instant alerts while browsing social media or other platforms, helping them stay safe in the digital space.

2. Problem Statement

Although many AI models for cyberbullying detection already exist, they come with limitations. Most of them are built only for English text, and they often fail when slang, short forms, Hinglish, or emojis are used. Moreover, there are very few tools that can work directly on the user's browser in **real time**.

Therefore, we aim to develop a tool that is not only accurate but also practical. The system should be able to **detect bullying content in English as well as Hinglish**, handle slang, and classify the seriousness of the bullying (mild, aggressive, or severe). This makes the internet a more respectful and secure place for communication.

3. Objectives

Our main goals in this project are:

1. To compare different algorithms (like SVM, Random Forest, LSTM, CNN, and Transformer-based models such as BERT, RoBERTa, mBERT) for detecting cyberbullying.
2. To find out which algorithm works best in terms of accuracy, speed, and overall performance.
3. To create a browser extension that can **highlight or flag bullying text instantly** while browsing.
4. To make the system more effective by adding **slang detection, Hinglish language support, and severity scoring**.
5. To design a simple and user-friendly interface so that anyone can use it without technical knowledge.

4. Scope

The scope of this project includes:

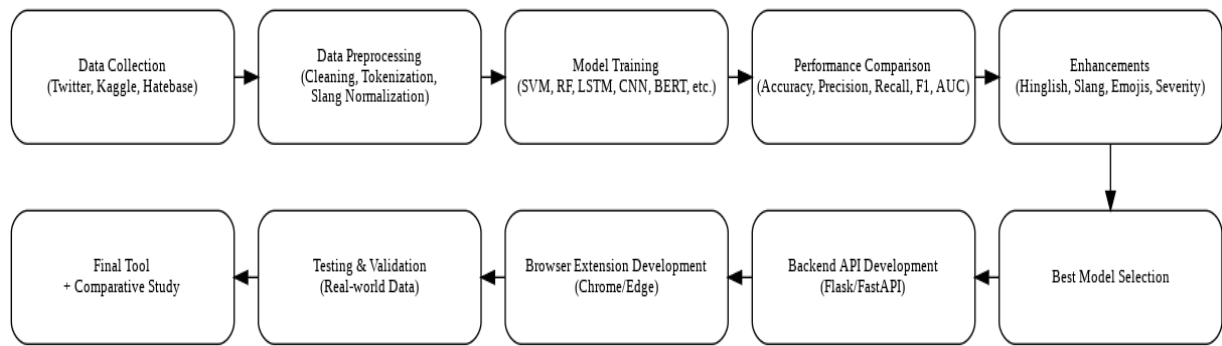
1. Comparing different algorithms such as SVM, Random Forest, LSTM, CNN, and Transformer-based models (BERT, RoBERTa, mBERT) to find the best one for cyberbullying detection.
2. Building a **browser extension** that can detect and highlight bullying text in real time while users browse social media or websites.
3. Supporting **English and Hinglish** text, along with slang, abbreviations, and emojis commonly used online.
4. Classifying cyberbullying into different levels such as **mild, aggressive, and severe**.
5. Providing both **research contribution** (comparative study of algorithms) and **practical utility** (a working tool for end users).

5. Brief Literature Survey

1. **Traditional Machine Learning methods** such as Support Vector Machine (SVM), Naïve Bayes, and Random Forest have been widely used with features like TF-IDF and Bag-of-Words.
2. **Deep Learning models** like LSTM, Bi-LSTM, and CNN have improved accuracy by learning word relationships through embeddings such as Word2Vec and GloVe.
3. **Transformer-based models** (BERT, RoBERTa, DistilBERT) have recently achieved state-of-the-art performance for text classification tasks.
4. **Multimodal approaches**, where text and memes are analyzed together using OCR + CNN, are also being explored.
5. **Existing tools** such as ReThink and StopIt provide some solutions, but they are either limited to English, or not directly usable as browser extensions.

6. Methodology

Workflow Diagram



We plan to develop the system in the following steps:

1. **Data Collection & Cleaning** – Gather cyberbullying datasets from Twitter, Kaggle, OLID, Hatebase, etc., and clean them by removing noise, slang normalization, and tokenization.
2. **Model Training** – Train and compare multiple algorithms like SVM, Random Forest, LSTM, CNN, and Transformer models.
3. **Performance Comparison** – Evaluate models using Accuracy, Precision, Recall, F1-Score, and ROC-AUC.
4. **Enhancements** – Add features like severity classification, Hinglish and emoji handling.
5. **Best Model Selection** – Choose the top-performing model(s) for deployment.
6. **Backend API Development** – Create a server using Flask/FastAPI to serve predictions from the model.
7. **Browser Extension Development** – Build a Chrome extension that sends webpage text to the backend API and highlights bullying content.
8. **Testing & Validation** – Test the tool with real-world data on different platforms to ensure accuracy and reliability.

7. Conclusion

By the end of the project, we expect to achieve:

1. A **comparative study of different algorithms** used for cyberbullying detection.
2. A **working browser extension** that detects and flags bullying text in real time.
3. Support for **English + Hinglish, slang, and emojis**.
4. Ability to classify cyberbullying into different severity levels (mild, aggressive, severe).
5. A practical tool that contributes towards making the internet a **safer and more respectful place** for its users.

8. Tools and Technologies

1. **Programming Languages:** Python, JavaScript
2. **Frameworks & Libraries:** TensorFlow/Keras, PyTorch, Scikit-learn, HuggingFace Transformers, Flask/FastAPI
3. **Browser Extension Development:** HTML, CSS, JavaScript, Chrome/Edge APIs
4. **Databases:** MongoDB / MySQL (if storage required for collecting new dataset)
5. **Other Tools:** GitHub, Google Colab/Jupyter, Cloud deployment tool

9. Project Plan (~24 Weeks)

Phase	Duration	Activities	Deliverables
Phase 1: Foundation	Week 1–4	Literature survey, dataset collection, preprocessing, baseline models	Initial results
Phase 2: Algorithm Experiments	Week 5–10	ML/DL/Transformer training & evaluation models on english dataset	Comparative analysis
Phase 3: Advanced Enhancements	Week 11–14	Hinglish support, slang dictionary, ensemble methods	Enhanced models
Phase 4: Extension Development	Week 15–20	API creation, extension prototype, UI integration	Working extension
Phase 5: Testing & Finalization	Week 21–24	Large-scale testing, optimization, documentation	Final tool + report

10. References

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